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eMap 2.0: A Web-Based Platform for Identifying electron Transfer Pathways in Proteins and Protein Families

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ABSTRACT

In this review we present eMap 2.0, a web-based application for predicting electron/hole transfer pathways in proteins and protein families based on their structures. The underlying model can be viewed as a coarse-grained version of the Pathways approach by Beratan and Onuchic [Beratan et al. J. Chem. Phys. 1987, 86, 4488]. Similar to the original framework, eMap employs graph-theory algorithms to search for the most efficient electron transfer pathways as shortest paths on a graph representation of the protein. In eMap, the nodes represent electron transfer active sites and only through-space tunneling is considered for each individual electron/hole hop. eMap 2.0 takes this model one step further by aiming at identifying shared electron transfer pathways in protein sets. From a graph theory standpoint, this is achieved using frequent subgraph mining (FSM) algorithms. Lastly, eMap 2.0 utilizes sequence and structural similarity measures to analyze and cluster the results. Here, we show how this robust method can be utilized to rapidly provide insights regarding conserved electron transfer pathways within protein families and to identify outliers, in which the conserved electron transfer pathway is blocked either by a mutation or conformational changes.

This article is categorized under:

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1 | Introduction

Biological processes, such as respiration [1, 2], photosynthesis [3, 4], and DNA photo-damage repair [5, 6] utilize electron or hole transfer. It has been shown that approximately one third of the structurally characterized proteins exhibit extensive chains of Trp and Tyr residues [7]. This observation led to the suggestion that these chains can have a protective function against oxidative damage by guiding potentially chemically reactive holes away from the active site towards the surface of the protein [7]. Indeed, Trp and Tyr have been shown to participate in electron/hole shuttling in various protein families [8] including but not limited to cryptochromes [9, 10], photolyases [6, 11], cytochrome c peroxidasse [12], blue light receptors [13], photosystem II [14], oxalate decarboxylases [15], and ribonucleotide reductases [16].

Obtaining a theoretical description of electron transfer in proteins that yields quantitative observables such as electron transfer rates is a challenging but often a feasible task provided that the key players are known either from experimental data (e.g., mutagenesis studies), or from preliminary simulations [16–19]. Detailed characterization of electron transfer requires accounting for the dynamic behavior of a protein and a careful description of nuclear and electronic degrees of freedom, especially for proton-coupled electron transfer events [20]. While this can be achieved for a single protein system when the participating residues and co-factors are known,

James R. Gayvert and Alyssa J. Kranc contributed equally to this study.

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rigorous quantitative characterization of electron transfer process without any a priori knowledge is seldom feasible computationally.

Several theoretical models have been proposed to enable blind screening for electron transfer pathways in proteins using the protein structure as an input. The Pathways model [21] proposed by Beratan and Onuchic describes electron transfer as a sequence of through-bond, through-space, and through-H-bond atom-to-atom tunneling events. Each electron hop is parameterized by an empirical penalty function. The model relies on graph theory algorithms to find the most efficient electron transfer pathway as a shortest path on a connected graph. Each protein atom represents a node in the graph, and the edge weights are specified by the penalty function [21]. The outcome of the simulation is a series of pathways ranked based on the predicted tunneling matrix element values. A more rigorous approach based on semi-empirical quantum-chemical calculations has been proposed by Stuchebrukhov and co-workers [22]. The model and the corresponding software, Electron Tunneling in Proteins program (ETP) [22], quantifies and visualizes electron tunneling current through a protein. A more recent EHPath model [23] by Beratan and co-workers targets electron/hole hopping through redox-active sites in the protein. EHPath identifies and ranks electron transfer pathways in the protein based on the pathway's mean residence time. EHPath relies on a kinetic model parametrized using Marcus rate constants for the electron hopping events and, thus, inevitably requires information on the Gibbs free energies, reorganization energies, and electronic couplings [23]. The eMap model [24], proposed earlier by us, parametrizes electron transfer pathways as a series of through-space electron tunneling events between electron transfer—active sites, which avoids an atomistic description of the process, but otherwise follows the Pathways model. As such, it optimizes the tunneling matrix element, or effective couplings between the donor and acceptor, along the path [25]. Therefore, the pathways predicted by eMap are better described as electron transfer coupling pathways, following the terminology used, for example, in Ru et al. [26]. Despite the constraints imposed by the model, eMap successfully predicts electron transfer pathways, including hopping pathways, in multiple protein systems, such as cryptochromes [24] and additional examples discussed below. The simplicity of the model allows for a quick screening for efficient electron transfer pathways based solely on the protein structure. The model has been implemented as a web application with an intuitive interface that is available at https://emap.bu.edu, and its backend component, PyeMap, is available on GitHub [27]. Here we describe the new features implemented in the current release of eMap 2.0, in particular the ability to screen series of proteins for shared electron transfer pathways. We discuss multiple modes of how one can use eMap 2.0 to analyze a single protein or protein families and present test cases which show the potential of the software to identify individual and shared electron transfer pathways.

2 | Model

The eMap web server provides a user-friendly interface enabling both single and multiple-protein analyses, while a fully-functional backend is also available as a stand-alone package, PyeMap [27]. In what follows, we use eMap 2.0 to refer to both the model and the software; all developments described below have also been implemented in the PyeMap package.

Following the Pathways model [21], eMap relies on graph theory to identify candidates for efficient electron transfer pathways in a protein. In eMap 2.0, we apply graph mining approaches to search for shared electron transfer pathways (or structural motifs) across a group of protein structures. Detailed definitions of graph mining terms that will be used in the following sections can be found in Section S1 of Supporting Information.

2.1 | Single Protein Analysis

The eMap model of a single protein can be viewed as a coarse-grained adaptation of the *Pathways* model [21, 25, 28, 29], where only through-space tunneling between aromatic residues and other user-specified sites is considered. In the original *Pathways* model, an electron/hole transfer pathway between a specified donor and acceptor is described as a series of through-space, through-covalent bond, and through-hydrogen bond tunneling events. The tunneling matrix element (T_{DA}), an effective donor-acceptor coupling, can be approximated as follows assuming multiplicative penalty functions ($\varepsilon_{i/j/k}$) for each tunneling event: [21]

$$T_{DA} = \text{const} \times \prod_{i \in space} \varepsilon_i \times \prod_{j \in bond} \varepsilon_j \times \prod_{k \in H-bond} \varepsilon_k \tag{1}$$

Thus the task of finding the most efficient pathways can be formulated as identifying those which maximize the matrix element T_{DA}.

In the eMap model, the protein structure (with one or more chains) is used to construct an undirected weighted *protein graph*, G, where each aromatic residue or a user-specified site represents a node, and each edge has a weight associated with a through-space tunneling penalty function:

$$\varepsilon = \alpha \exp\left[-\beta\left(R - R_{\text{offset}}\right)\right] \tag{2}$$

where α, β, and R_{offset} are empirical parameters, and R is the distance between the sites defined either as the distance between the centers of mass of the sites or as the distance between the closest atoms. To make use of standard graph theory methods, it is convenient to operate with non-negative edge weights and additive parameters rather than multiplicative penalty functions. Following the original *Pathways* model, this is done using a modified penalty function as the edge weight:

$$P' = -\log \varepsilon \tag{3}$$

where ε is the original penalty function (Equation 2). With this parametrization, the edge weights become scaled distances between the hopping sites (α=1). Since maximizing the product of the penalty functions ε is equivalent to minimizing the sum of the modified functions P' and since the modified penalty function is proportional to the distance between the sites (Equation 3), finding the path with the largest T_{DA} value is equivalent to finding the shortest path between the given donor and acceptor in the graph [21, 25, 28, 29]. For graphs with non-negative weights, this problem is easily solved using Dijkstra's algorithm [30] to find

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the optimal path, or Yen's algorithm [31] to find deviations from the optimal path connecting given source and target.

The default values of the parameters are as follows: α (=1.0), β (=2.3 Å^{-1}), and R_{offset} (=0.0 Å). To reduce the complexity of the protein graphs, and therefore provide a more meaningful network structure, the edges of the graph are pruned, favoring those with the smallest weights [32]. Lastly, nodes in the graph are classified as buried or surface-exposed using either the residue depth [33–35] or relative solvent accessibility [36–38] criteria. Details of the implementation can be found in Refs. [24, 32].

eMap 2.0 introduces only minor changes in the single-protein analysis routine, such as an expanded set of automatically identified redox-active sites. The workflow of the eMap single protein analysis is illustrated in Figure 1. The user can use PDB ID or structure files (PDB or CIF) as an input for the software. The structure is then parsed, and redox-active sites are identified. By default, aromatic side-chains of Tyr and Trp are considered active along with aromatic fragments of co-factors. Residues such as Trp and Tyr have been demonstrated to participate in hopping transport in systems such as ribonucleotide reductases and cryptochromes/photolyases [8]. In addition to aromatic fragments, redox-active metals and iron–sulfur clusters are automatically identified and can be included in the analysis as redox-active sites. In the next step, a graph representing the protein structure is generated. The user then specifies an electron/hole donor and optionally an acceptor, and the software searches for the shortest paths connecting the donor and acceptor or donor and all residues on the protein surface. The shortest pathways are visualized on a graph (in 2D) and in 3D using NGL viewer [39, 40].

eMap 2.0 naturally inherits all the advantages and limitations of the original model for single protein analysis. Among the main advantages are the simplicity and quick predictions. One of the key assumptions of the model that enables computational simplicity is that the hopping probability is purely distance-dependent [24]. The downside of this assumption, in addition to lacking energetic information, is its implication for the protein graph analysis. The most efficient electron transfer pathway in the model corresponds to the shortest path on a connected graph between the donor and acceptor. Provided that the edge length is not capped, there can be an edge directly connecting the donor and acceptor regardless of the actual distance between the sites. Thus, the direct tunneling will be the preferred pathway predicted by the model. To avoid this, all edges longer than 20 Å are disregarded by default. One might consider an even tighter cutoff if the distance between the closest atoms is used to define the edge weight. Other parameters controlling the graph generation, such as edge pruning parameters, should also be chosen with care. However, despite the limitations, the eMap model has shown promising results for several classes of proteins it has been tested on, including the results presented in the current work and earlier [24].

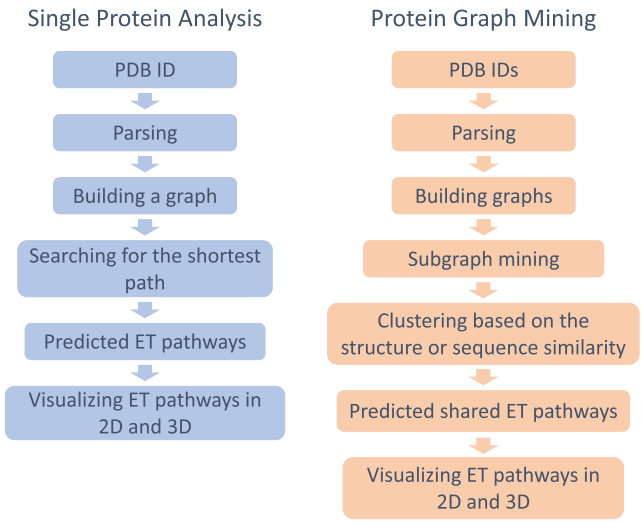


FIGURE 1 | The workflow of the eMap single protein analysis (left panel) and common subgraph mining for a series of proteins (right panel).

2.2 | Searching for Common Electron/Hole Pathways

Single protein analysis results in a protein graph for the protein of interest, and each pathway identified by eMap can be viewed as a subgraph of the protein graph. Therefore, using the eMap model, the task of identifying pathways (and other structural motifs) which are shared among a group of proteins can be formulated as identifying frequently occurring subgraphs in a data set of protein graphs. This is a well-studied problem in graph theory known as frequent subgraph mining (FSM) [41, 42].

FSM techniques are aimed at the discovery of patterns in graphs that exhibit a particular structure that are deemed interesting or relevant [43]. What makes a pattern "interesting" or not depends on the application. FSM techniques are useful where there are complex relationships between data entities and are commonly used in areas such as analysis of XML documents and web usage [44]. In the scientific community, FSM has seen numerous applications in bio- and cheminformatics contexts, including drug discovery [45], identification of protein-ligand interfaces [46, 47] and the study of evolutionary relationships through mining phylogenetic trees [48–50]. While others have used FSM techniques to identify conserved three-dimensional structures or motifs in proteins [46, 51–53], to our knowledge, eMap is the first to do so with an emphasis on electron transfer.

The FSM problem has a large degree of computational complexity. The most basic task, determining whether a graph G contains a pattern P, is an instance of the subgraph isomorphism problem, which is NP-complete [54]. Further, searching over all possible patterns P becomes rapidly intractable for large and dense graph datasets. Several approaches have been developed to efficiently tackle this problem, where the goal is typically to identify patterns which occur more often than a given threshold [43]. Here, we use the gSpan algorithm, implemented in the gspan-mining Python package [54–56]. gSpan is a complete algorithm, which identifies all subgraphs P that satisfy a minimum support threshold. For a graph dataset, $D = \{G_1, G_2, \ldots, G_n\}$, the support of P is the number of graphs $G \in D$ for which P is a subgraph. The details of the algorithm can be found in Ref. [54], and the key features which relate to its usage in the eMap software are discussed in Section S2 of Supporting Information.

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The following outlines the main steps of the eMap multiprotein analysis workflow.

Frequent subgraph mining (FSM) in eMap can be divided into three major steps: (i) graph generation; (ii) mining and matching; and (iii) clustering of identified subgraphs (Figure 2) [43]. Clustering of identified subgraphs is essential to our task as for a given frequent subgraph pattern P, which only represents connectivity, there can be hundreds of matches, and thus a meaningful analysis requires grouping them based on some metric of relevance. Since the objective of eMap is to identify shared pathways, we perform a clustering analysis to group protein subgraphs by their similarity. eMap employs two metrics of similarity: structural similarity, which is based on a modified root mean squared deviation (RMSD) between residues, and sequence similarity, which is based on a multiple sequence alignment performed using the MUSCLE package [57]. The mining and clustering stages are discussed in more detail in Sections S2 and S3 of Supporting Information.

3 | Application Examples

Below we discuss several different modes of how the eMap 2.0 software can be used to explore electron transfer pathways in a single protein and protein families. For multiple protein analysis, we present two test cases. The first corresponds to a blind search, where no information on the shared pathway is given as an input, and the second is an illustration of a search for a shared electron transfer pathway between a user specified donor and acceptor.

3.1 | Efficient Electron Transfer Pathways in a Single Protein

In earlier work, we described the application of eMap to identify an electron-transfer channel in a single cryptochrome protein [24]. Here, we present a case study of radical transfer pathway in class Ia ribonucleotide reductase (RNR) from *Escherichia coli*. RNRs catalytically reduce ribonucleotides into deoxyribonucleotides and, thus, play a key role in DNA biosynthesis [58]. Each class of RNRs approaches the requisite creation of the thiyl radical at the active site differently. Class Ia RNRs generate this thiyl radical by a long-range radical transfer from the diferric-tyrosyl radical co-factor in a neighboring protein chain [59–61](Figure 3). The electron transfer pathway involves six residues from the neighboring α and β subunits: βY122–βW48–βY356–αY731–αY730–αC439 [63–67].

Below we discuss the results of the eMap analysis for the protein using the cryo-electron microscopy (cryo-EM) structure of the active $\alpha 2\beta 2$ complex (PDB ID: 6W4X) [62] as the input.

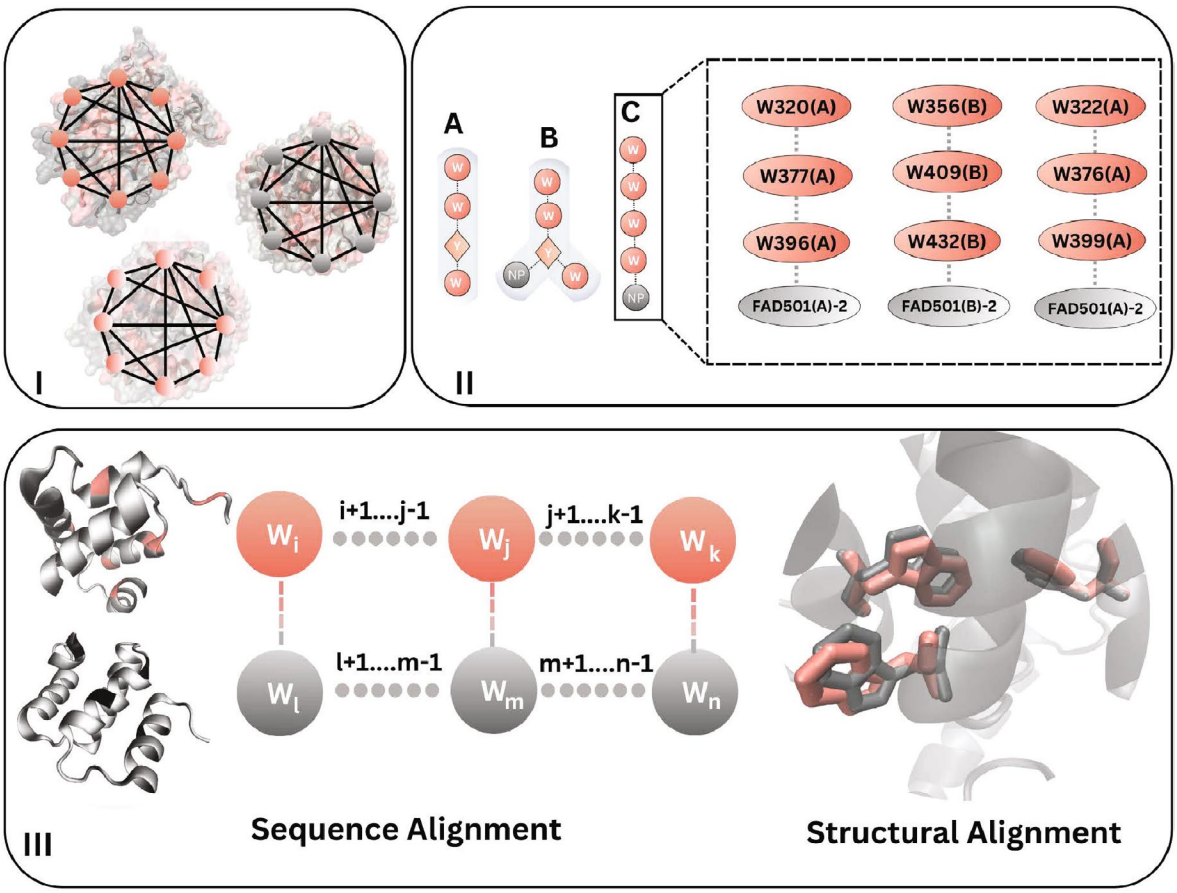


FIGURE 2 | Flowchart illustrating the three major steps of FSM with eMap. The first step (I) generates a graph data set which consists of protein graphs created by the single protein version of eMap. In the second step (II), the graph data set is mined for frequently occurring subgraph patterns. For each identified pattern, all instances of said pattern within the data set are found using graph matching, which yields a set of *protein subgraphs*. In the final step (III), the protein subgraphs matching a selected pattern are clustered into groups based on sequence or structural similarity.

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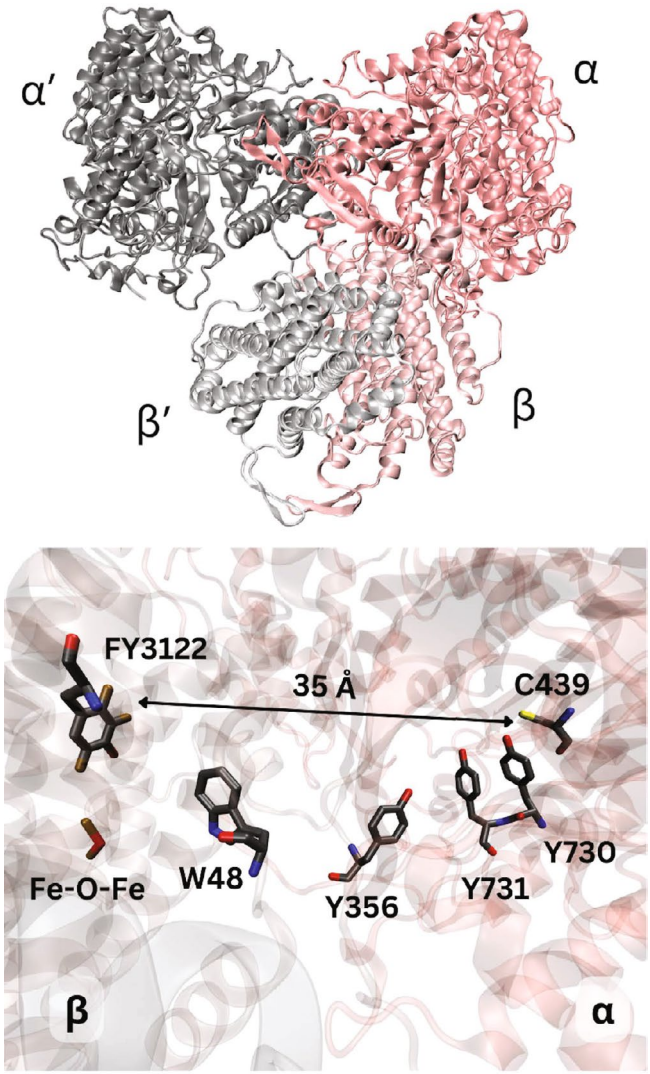


FIGURE 3 | Structure of the class Ia RNR from *E. coli* (top) and inter-unit radical transfer pathways involved in the catalysis, *β*Y122–*β*W48–*β*Y356–*α*Y731–*α*Y730–*α*C439 (bottom). PDB ID: 6W4X [62].

As shown in Figure 4a, the software detects aromatic 3-fluoro tyrosine (FY3) as well as aromatic moieties of co-factors: thymidine triphosphate (TTP, not shown) and guanosine diphosphate (GDP). Including Y, W, C, and all non-amino acid aromatic sites into the analysis and considering only chains B (*α*) and C (*β*) yields the graph shown in Figure 4a. Specifying FY3, that is, FY3122(C), as a source and C439(B) as a target of radical character transfer, and using default simulation parameters and the closest atom definition of the distance, the experimentally observed pathway, shown in Figure 4b is the top ranked pathway (Table 1), that is, is predicted to be the most efficient electron transfer pathway channel.

One of the fundamental limitations of the current eMap model is that it disregards the electron transfer energetic parameters. To explore the effects of incorporating the information on the redox free energies, reorganization energies, and electronic couplings, we compared the results generated with eMap to those produced by EHPath [23] using another model protein system, cytochrome P450 [23]. Both models yield the same pathway as the most efficient, quantified by the pathway score (eMap) and the pathway mean residence time (EHPath)–see SI for more details. While the agreement might not be universal, the results are encouraging. Moreover, the simplicity of the eMap model enables (i) its general applicability including cases when the energetic parameters for the electron-transfer active moieties and co-factors are not available, and (ii) its application in a high-throughput manner for screening large sets of proteins owing to its low computational cost.

3.2 | Shared Electron Transfer Pathways

A key feature added in eMap 2.0 is screening a group of PDB files for the existence of common electron transfer pathways. There are two primary ways of using eMap to explore electron transfer in protein families. The first is a blind search for shared pathways in a series of proteins. The other is a guided search when some mechanistic information on the participating residues is available. Below we present case studies illustrating these two applications.

3.2.1 | Blind Search: Cryptochromes and Photolyases

To illustrate how eMap 2.0 can be applied to identify shared electron transfer pathways using graph mining tools we have first considered a series of 27 flavoprotein structures: 1U3D [68], 1U3C [68], 6LZ3 [69], 2J4D [70], 6FN2 [71], 1NP7 [72], 3ZXS [73], 4GU5 [74], 4I6G [75], 6PU0 [76], 1IQR [77], 4U63 [78], 6KII [79], 3FY4 [80], 1DNP [81], 1QNF [82], 1IQU [77], 2WB2 [83], 6RKF [84], 1O96 [85], 1EFP [86], 1O97 [85], 1X0P [87], 2Z6C [88], 1G28 [89], 4EER [90], and 2IYG [91] (see Table S3 in Supporting Information for more details). The set includes 7 plant cryptochromes, 3 animal cryptochromes, 8 photolyases, and 9 non-cryptochrome and non-photolyases flavoproteins. Light absorption by cryptochromes and photolyases initiates photo-induced electron transfer involving the FAD cofactor and conserved Trp residues [92–95]. Plant cryptochromes electron-transfer pathway involves three conserved Trp residues, referred to as the Trp triad [92, 95]. The electron transfer pathway in animal and animal-like cryptochromes is extended by a fourth aromatic residue (Trp or Tyr) [96]. To test whether eMap can identify the shared electron transfer pathway in cryptochromes and photolyases, we performed a blind search for all frequent subgraphs, that is, no input was given to the software on the nature of the electron transfer pathway. The search was performed using the default parameters and a minimum support of 18. The mining yielded 25 shared subgraph patterns with support of at least 18. These patterns themselves, however, do not necessarily correspond to shared electron transfer pathways. They merely represent the connectivity of the nodes. Thus, the same pattern can be associated with entirely distinct arrangements of the residues within protein structures. To further determine whether a found pattern represents a conserved electron transfer pathway, all occurrences of the pattern in the protein structures were clustered using structural similarity criteria. Indeed, only five of the 25 found shared subgraph patterns contained subgraph groups with significant structural similarity (Figure 5). The established electron transfer pathway corresponds to the major cluster identified for patterns 21 and

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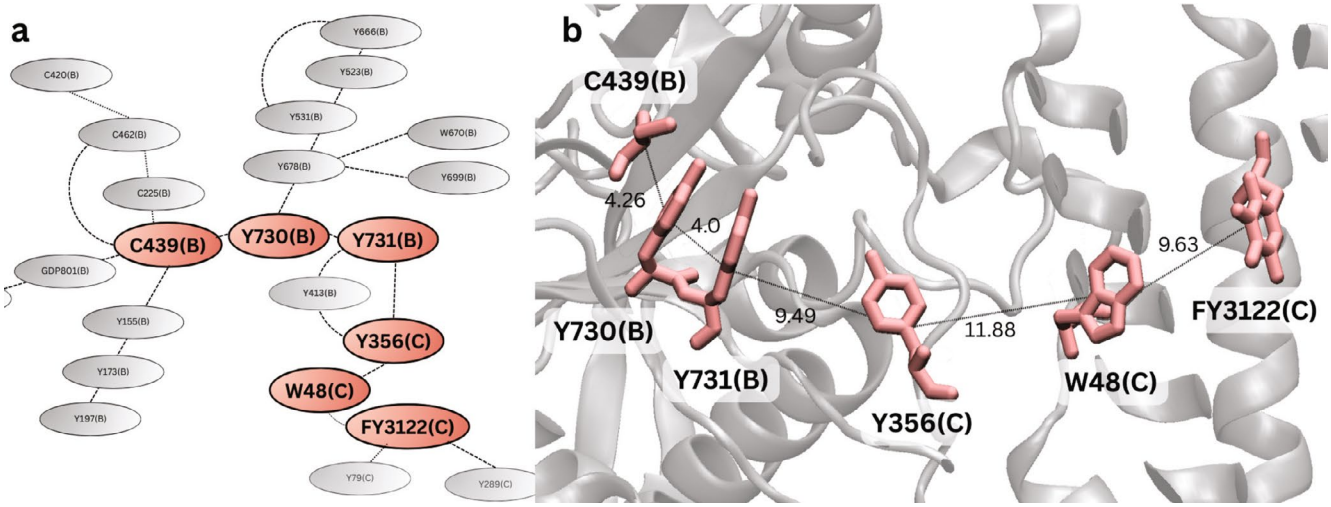


FIGURE 4 | The results of the eMap analysis for the 6W4X structure. Only B and C chains were included in the analysis, and all Y, W, and C residues were considered as ETA sites. All aromatic moieties of co-factors and non-standard amino acids were also considered as ETA sites. The default parameters used for the simulation are provided in Table S2 in the Supporting Information. The predicted radical transfer pathway for Ia RNR from *E. coli* is shown as a graph in panel (a) and in 3D in panel (b). The path coincides with the established inter-unit radical transfer pathway: βY122–βW48–βY356–αY731–αY730–αC439 [63–67]. Distances between the centers of mass of the amino acid residues participating in the electron transfer are reported in Å.

TABLE 1 | Four identified electron transfer pathways from FY3122 (β subunit) to C439 (α subunit) in class Ia RNR from *E. coli*.

Source	Node 1	Node 2	Node 3	Node 4	Node 5	Node 6	Score
FY3122(C)	W48(C)	Y356(C)	Y731(B)	Y730(B)	C439(B)		39.26
FY3122(C)	W48(C)	Y356(C)	Y413(B)	Y730(B)	C439(B)		41.23
FY3122(C)	W48(C)	Y356(C)	Y731(B)	Y413(B)	Y730(B)	C439(B)	45.83
FY3122(C)	W48(C)	Y356(C)	Y413(B)	Y731(B)	Y730(B)	C439(B)	46.61

Note: The Score (column 8) is the sum of edge weights in the path, which with the default penalty function parameters is approximately equal to the sum of the distances between the nodes along the path. The smallest score corresponds to the length of the shortest path in Å.

22 in Figure 5. This pathway is found by eMap in 16 proteins, those that are either cryptochromes or photolyases. The only two cryptochrome structures which do not exhibit this pathway are 6LZ3 and 3ZXS. 6LZ3 is a cryo-EM structure of a tetramer of CRY1c from *Zea mays* with the second Trp in the Trp-triad replaced by Ala (W368A mutation following the residue indexing in 6LZ3) [69]. 3ZXS is an X-ray structure of CRYb from *Rhodobacter Sphaeroides* that contains Tyr (Y387) in place of the first Trp in the triad [73], while the retained Trp residues are in fact structurally different from the canonical Trp-triad of cryptochromes/photolyases. This structure falls within a distinct CryPro (protobacterial cryptochromes) sub-cluster of the cryptochrome/photolyase family [73]. Thus, the eMap software correctly identified all proteins (cryptochromes and photolyases) that exhibit the conserved Trp triad.

Interestingly, some of the other shared subgraphs identified by eMap could also have mechanistic importance. Subgraph patterns 8 and 9 include adenine and flavin moieties of the FAD cofactor as well as three Trp residues. Two of these Trp residues belong to the conserved Trp triad (W420 and W397 in PDB ID 4GU5 [74], see Figure 6), while the third Trp (W353) is also highly conserved among the 18 considered photolyases and cryptochromes and occurs in 16 of them.

Importantly, the remaining subgraph patterns only contain clusters with eight or fewer proteins, which indicates that they are unlikely to correspond to mechanistic pathways common for cryptochrome or photolyases families of proteins.

3.2.2 | Guided Search: Heme-to-Heme Electron Transfer in Bacterial Cytochrome c Peroximdases

To demonstrate eMap's ability to identify shared electron transfer pathways given some input information on the participating residues, we considered the bacterial cytochrome c peroxidase (bCcPs) family of proteins, which catalyze the reduction of hydrogen peroxide and are part of the oxidative stress protection machinery of many Gram-negative prokaryotes [97].

The catalytic cycles of canonical bCcPs relies on electron transfer between the *heme* sites. Trp94 is thought to mediate electron shuttling between the two heme centers [97, 98]. To test whether eMap analysis supports this channel, we performed a search of Heme–⁎–Heme pathways, where '⁎' indicates a wild card character, that is, any residue. We considered a subset of 11 structures belonging to the bCcP superfamily that contain two hemes: 6V59 [99], 6NX0 [100], 1EB7 [101], 2VHD [102], 1RZ5

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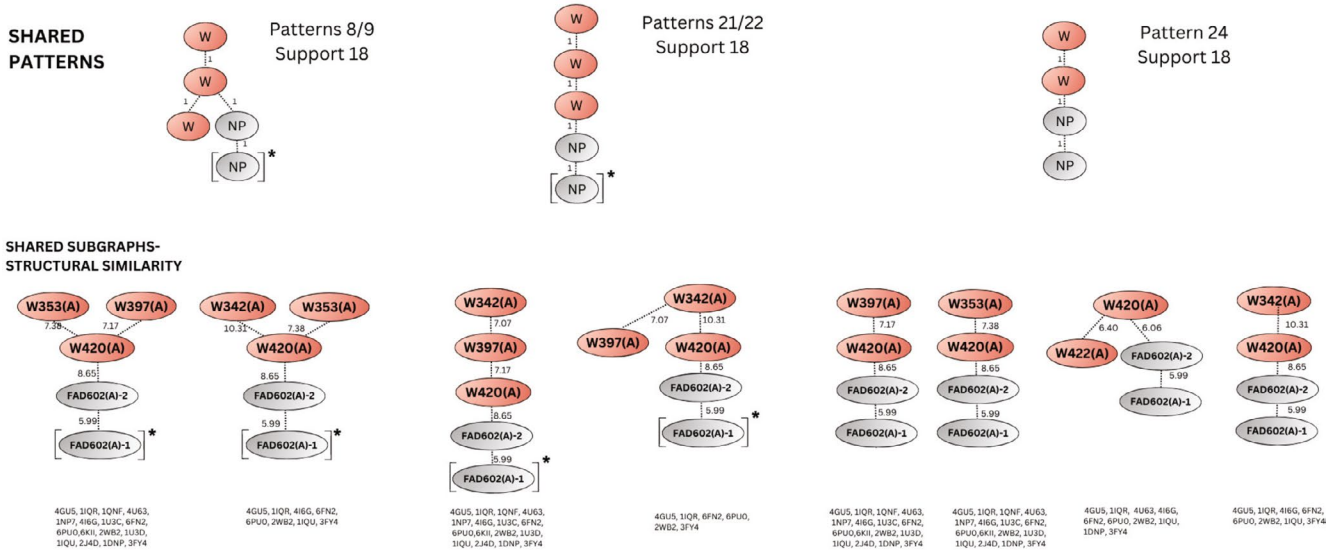


FIGURE 5 | Relevant shared graph patterns and subgraphs for a test set of 27 flavoproteins, containing cryptochromes, photolyases, and non-cryptochromes and non-photolyases flavoproteins. Only the shared patterns with clusters containing more than 15 proteins based on structural similarity criteria are shown. Groups 8/9 and 21/22 only differ by presence (8 and 21) or absence (9 and 22) of adenine of FAD co-factor, which is represented by []* in the figure. The residue IDs correspond to Drosophila Cryptochrome, PDB ID 4GU5. FAD602(A)-1 and FAD602(A)-2 stand for adenine and flavin fragments of the FAD co-factor, respectively.

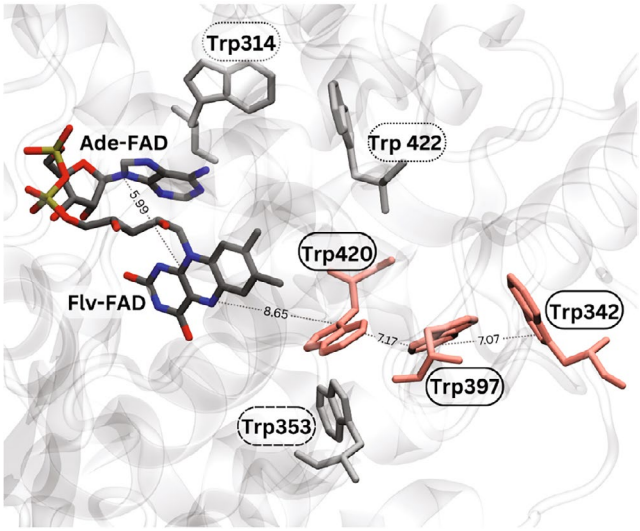


FIGURE 6 | Residues involved in photo-induced electron transfer: Canonical Trp triad (Trp420, Trp397, and Trp342) as well as other Trp residues from shared subgraphs identified by eMap. The numbering and structure correspond to PDB ID 4GU5. Distances between the centers of masses of the relevant ET-active moieties are shown in Å.

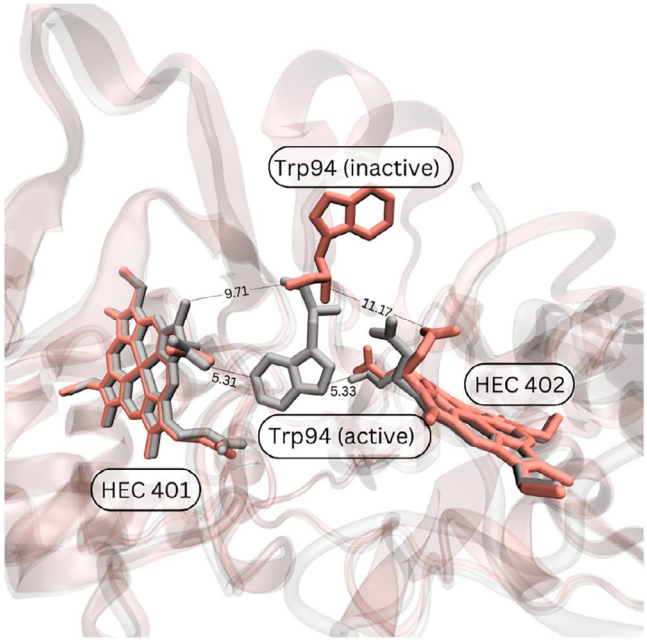


FIGURE 7 | Active and inactive forms of chain A in the di-haem Cytochrome C Peroxidase in *Marinobacter nauticus* (PDB structures: 1RZ5, 1RZ6). Shown are overlays of the active and inactive system, where Trp94 is shifted out of the central location between hemes in the inactive form. This inactive structure was not identified as a Heme-Trp-Heme pathway by eMap. HEC is shorthand for Heme C. Closest-atom distances are reported in Å.

[103], 2C1U [104], 2C1V [104], 1IQC [105], 1RZ6 [103], 1NML [103], and 1ZZH [106] (see Table S4 in Supporting Information for more details). The search was performed using the closest-atom distance criteria between residues, and the default settings were used for all other parameters (Table S2). The Heme-Trp-Heme path was found in seven structures: 1EB7, 2VHD, 1RZ5, 2C1U, 2C1V, 1IQC, and 1ZZH. This pathway was correctly not identified in the other four structures screened. In the cases of 6V59 and 6NX0, both BthA enzyme structures, a Ser residue rather than a Trp is found in-between the two Hemes. While canonical bCcPs use a Trp in this position to facilitate electron transfer, BthA does not [98, 100]. Consequently, the Heme-Trp-Heme pathway is not found. 1RZ6, 1NML, and 1RZ5 are especially interesting cases because they are all *Marinobacter* structures, but the pathway was only identified in the latter by eMap. The absence of the Heme-Trp-Heme pathway in the 1RZ6 and 1NML structures is explained by the fact that neither

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of these structures were crystallized in the active form, and the protein loop containing the Trp residue is shifted out of alignment to a position where it is no longer directly located between the donor and acceptor hemes [103] (Figure 7). The ability of eMap to make this distinction between inactive and active redox pathways highlights the descriptive power of the relatively simple underlying model of electron transfer.

The successful examples described above lay the groundwork for screening large protein sets for conserved electron transfer pathways. However, high-throughput analyses would be hampered by limited structural data. This can be overcome, for example, by using structural modeling tools, such as AlphaFold [107, 108]. Thus, eMap 2.0 together with bioinformatic and structural modeling tools can be used to analyze the vast, yet unexplored, sequence data to identify proteins with novel functions and new proteins that exploit mechanistically established electron transfer pathways.

4 | Conclusions

The eMap 2.0 web application is a user-friendly tool which enables prediction and visualization of electron and hole hopping pathways in a single protein, and identification of shared pathways in families of proteins. This robust and simple model only uses structural information as input. Here we have reviewed different modes of using eMap: from a single protein analysis to blind and guided searches for shared electron transfer pathways in protein sets. For single protein analysis, we have shown that the model correctly identifies the established electron transfer pathway in a class Ia RNR protein as the most efficient one. For families of proteins, we have demonstrated the capability of the software to identify shared pathways in protein families considering sets of flavoproteins and bCcPs as examples. The software is publicly available at https://emap.bu.edu free of charge, and the open-source backend PyeMap is available on GitHub [27].

Author Contributions

James R. Gayvert: data curation (equal), formal analysis (equal), investigation (equal), methodology (equal), software (lead), validation (equal), visualization (equal), writing – original draft (equal), writing – review and editing (equal). **Alyssa J. Kranc:** data curation (equal), formal analysis (equal), investigation (equal), methodology (equal), software (equal), validation (equal), visualization (lead), writing – original draft (equal), writing – review and editing (equal). **Ruslan N. Tazhigulov:** formal analysis (equal), investigation (equal), methodology (equal), software (equal), validation (equal), writing – original draft (supporting), writing – review and editing (supporting). **Ksenia B. Bravaya:** formal analysis (equal), investigation (equal), methodology (equal), visualization (supporting), writing – original draft (equal), writing – review and editing (equal).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The open-source software is available on GitHub. All information required to fully reproduce the reported data is provided in the main text and Supporting Information.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.

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